Slides and example codes are available:
https://github.com/mint-lab/cv_tutorial

Color

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Photometric Image Formation

- **Camera**: A light measuring device
  1. A **light source** emits light.
  2. The light is reflected or absorbed or passes through (e.g. transparent) an **object**.
  3. The (reflected) light goes through the **camera lens** and hits the **image sensor**.

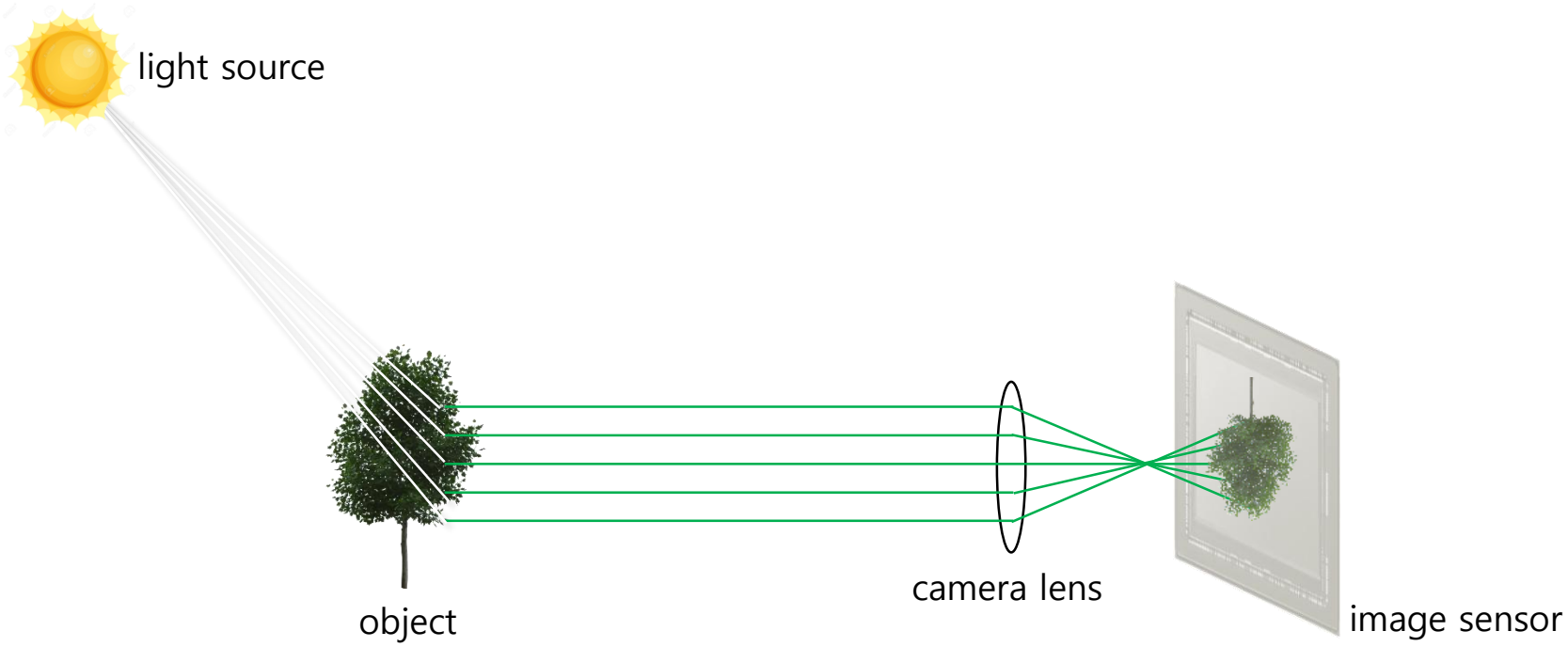


Image: PNG EGG, 123RF, PetaPixel

Light

- The **intensity** of light is the **amount of light energy** per unit area per unit time. (unit: W/m^2 or lux)
- The **color** of light is characterized by the **frequency** (unit: Hz) or **wavelength** (unit: m).

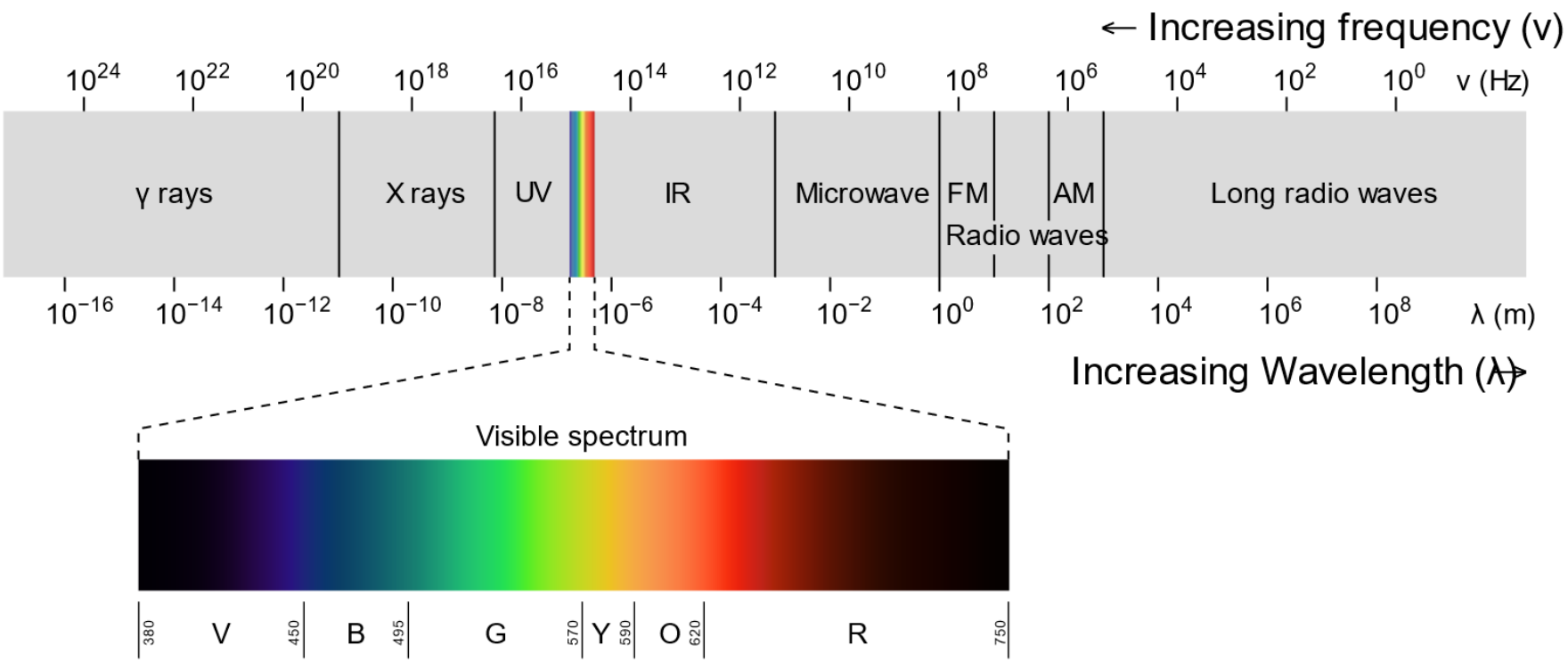


Image: Wikipedia

Photometric Image Formation

- CMOS sensors (or CCD sensors) are an array of photodetectors that can detect the intensity of light.

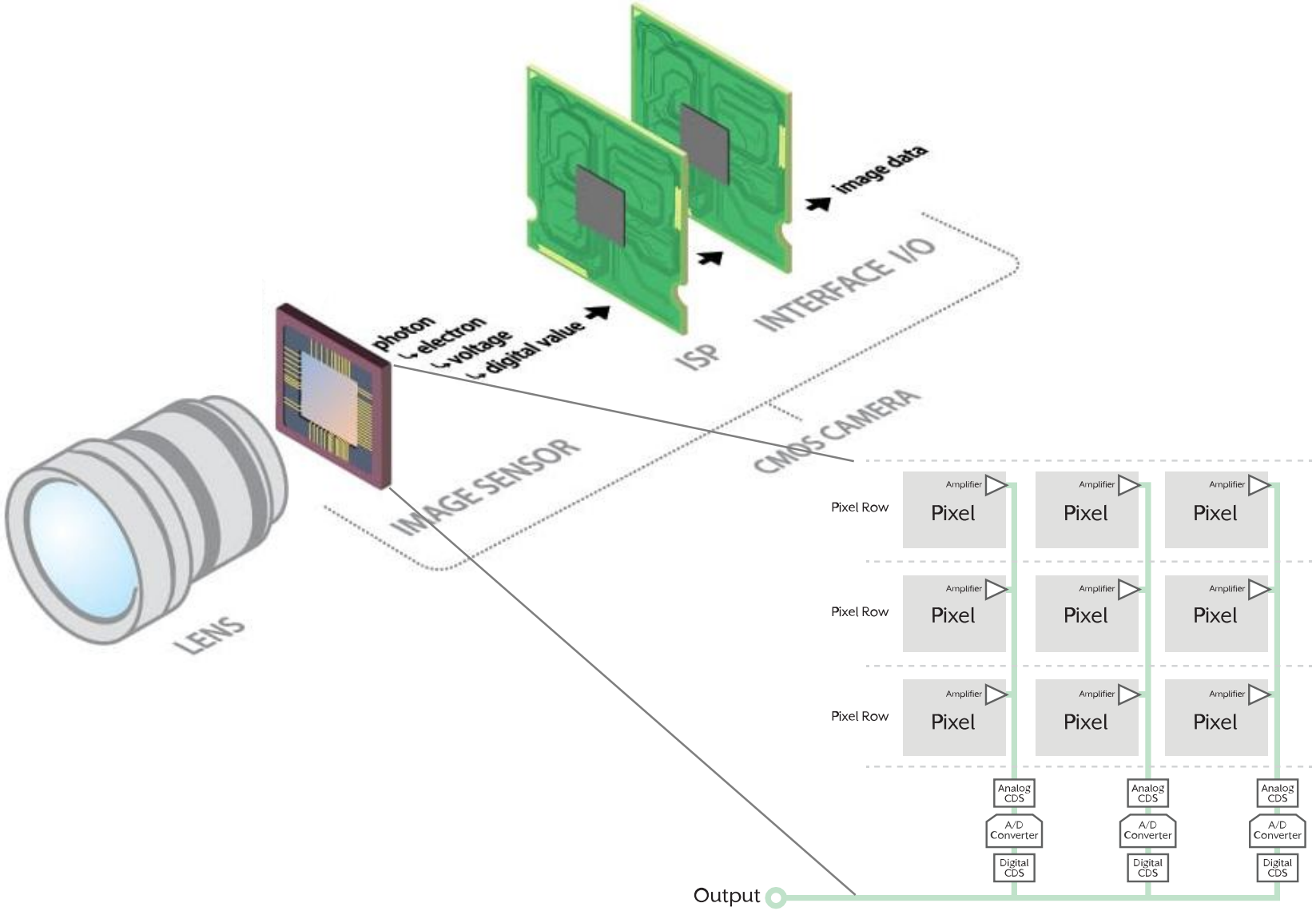


Image: LUCID Vision Labs

Photometric Image Formation

- CMOS sensors (or CCD sensors) are an array of photodetectors that can detect the intensity of light.
- Q) How does a camera control the intensity of light?
  - Aperture (조리개 in Korean)
    - A opening in a lens
    - Unit: f-number (f/N; N is f-number)
  - Shutter speed (a.k.a. exposure time; 노출시간 in Korean)
    - The length of time that an image sensor is exposed to light
    - Unit: Second
    - Note) Global shutter vs. Rolling shutter
  - ~~ISO sensitivity (a.k.a. exposure index)~~
  - Additional light sources (e.g. flash)

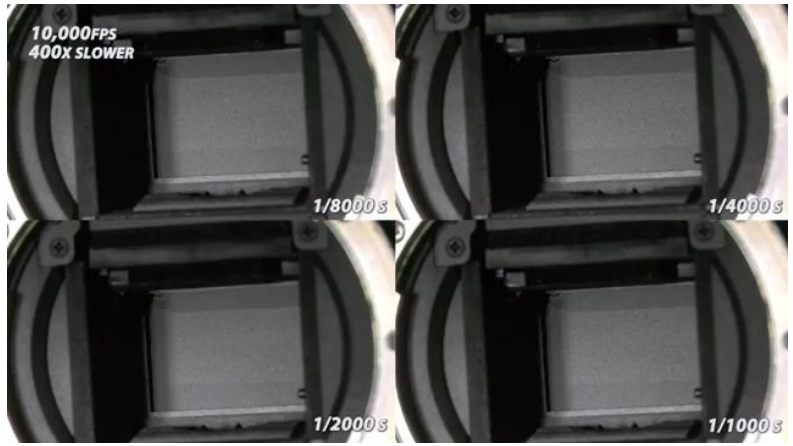
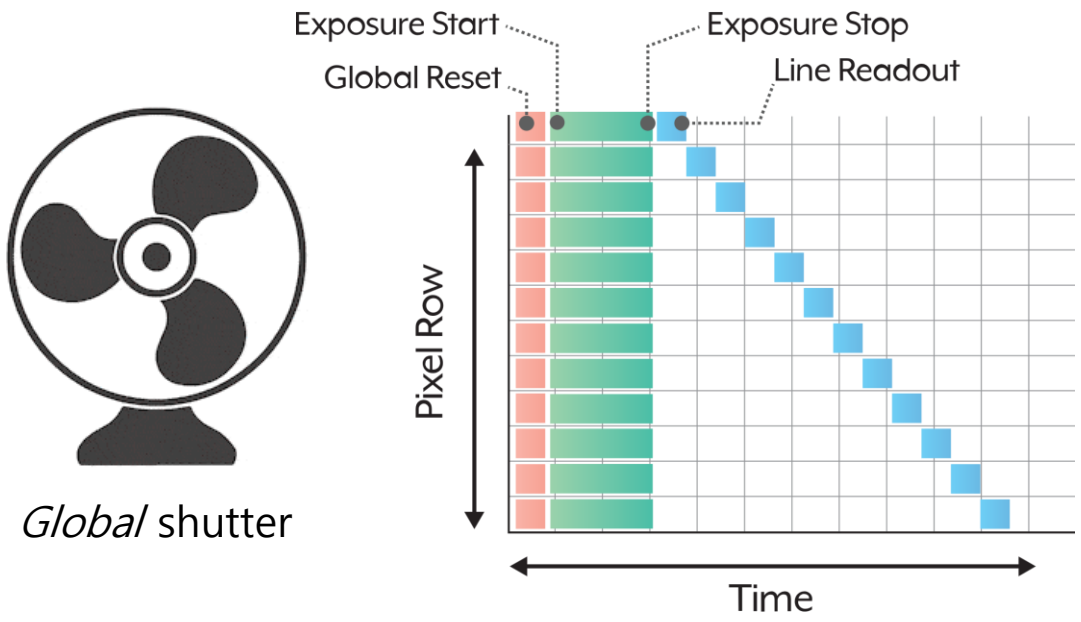


Image: Wikipedia and YouTube

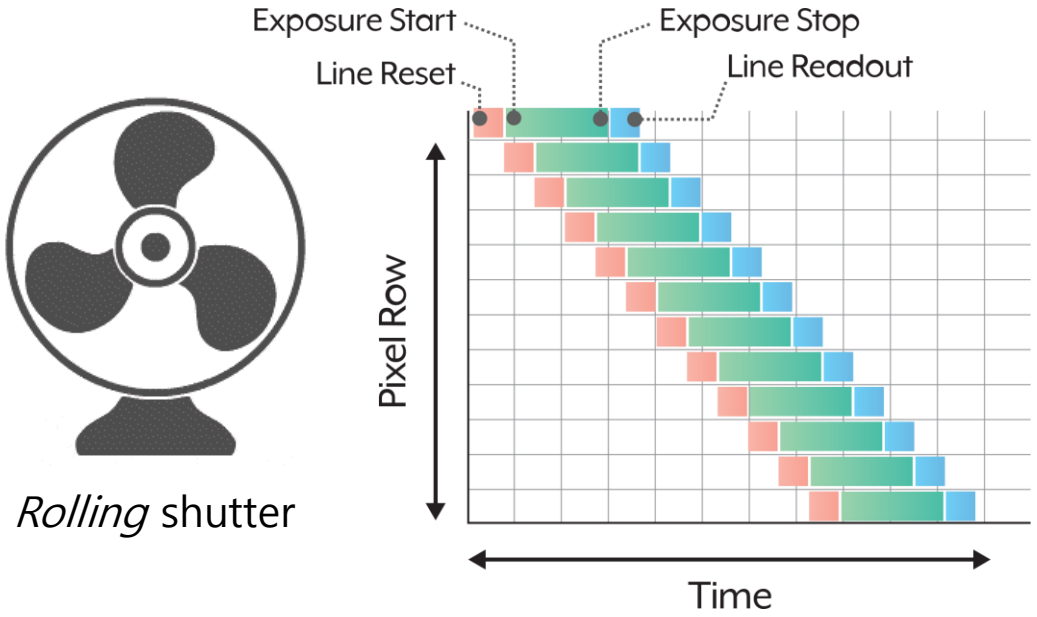
Photometric Image Formation

- CMOS sensors (or CCD sensors) are an array of photodetectors that can detect the intensity of light.
- Q) How does a camera control the intensity of light?
  - Shutter speed (a.k.a. exposure time; 노출시간 in Korean)
    - The length of time that an image sensor is exposed to light
    - Unit: Second
    - Note) *Global* shutter vs. *Rolling* shutter



Global shutter

vs.

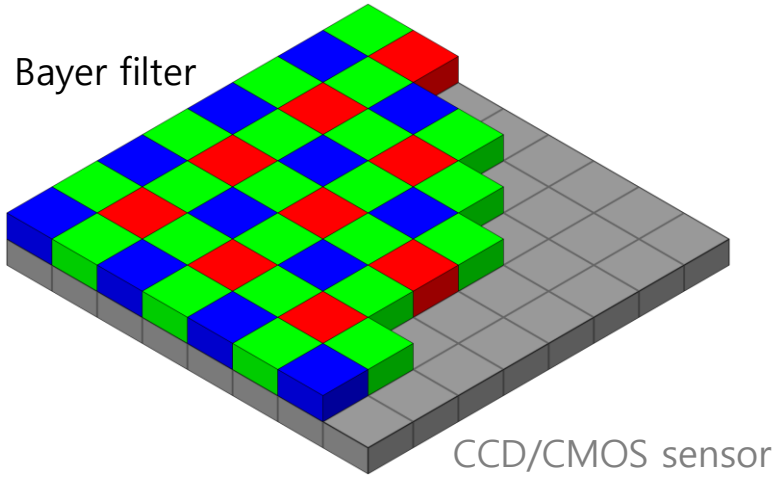


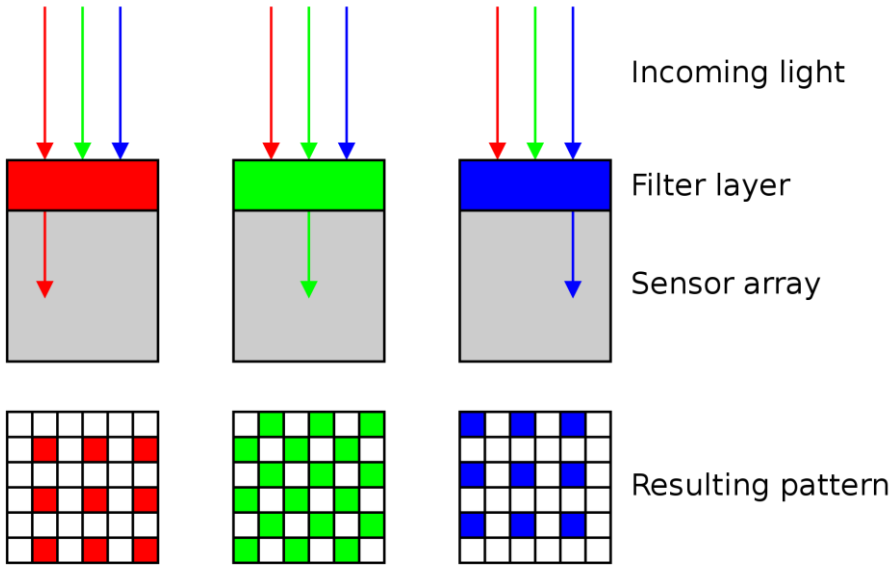
Rolling shutter

Image: LUCID Vision Labs

Photometric Image Formation

- CMOS sensors (or CCD sensors) are an array of photodetectors that can detect the intensity of light.
- Q) How does a camera capture (or distinguish) the color of light?
  - Color filter array for RGB separation
    - e.g. A Bayer filter is a the most common color filter array for RGB imaging.
  - A demosaicing algorithm (inside of a camera) reconstructs RGB images (④) from Bayer-sampled raw images (③).





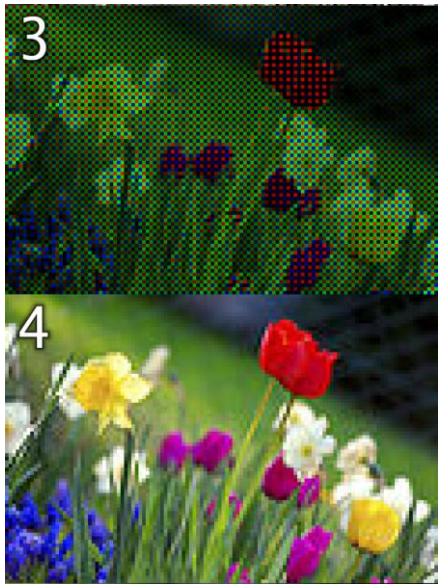


Image: Wikipedia

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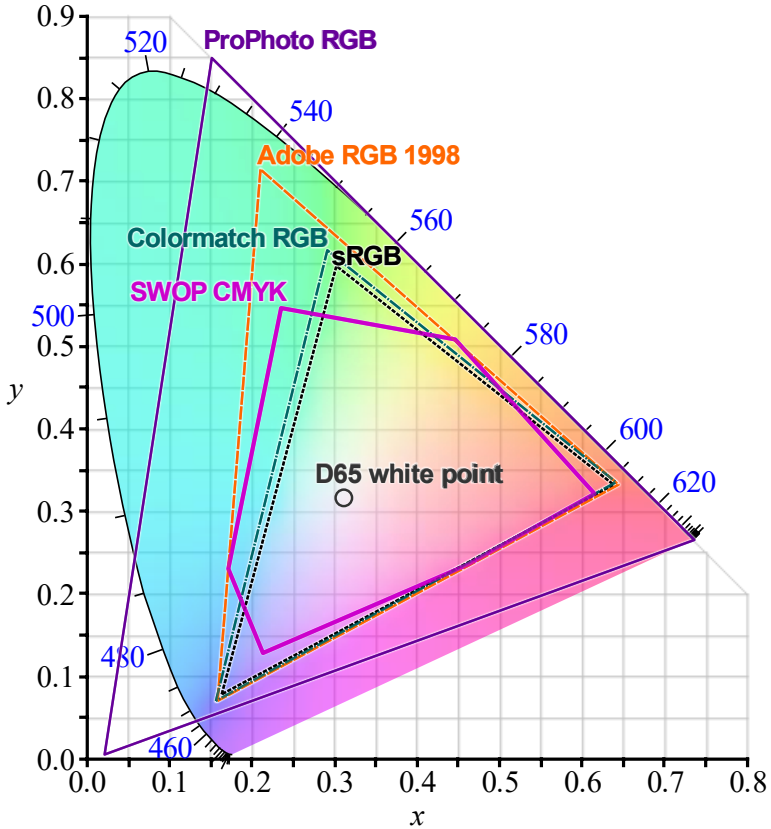
  - Color spaces for **human-friendly color selection**
  - Color spaces for **human visual perception**
- **Color Image Processing**
  - Color balancing
  - Color segmentation
  - Color histogram
  - How to use algorithms only support gray-scale images
    - Example) Color histogram equalization

Color Spaces

- A **color space** is a specific system for representing color numerically and graphically. [show its list]
- Color spaces for **media (e.g. display, printing, and broadcasting)**
  - **RGB**: An additive color space for displays (emitting light; e.g. television)
    - **sRGB**: A standard RGB color (IEC 61966-2-1:1999)
      used in monitors, printers, and the web
  - **CMY**(K): A subtractive color space for printing on white papers
  - Note) Chromaticity diagram (→)
    - sRGB: A monitor
    - SWOP CMYK: A printer
    - ProPhoto RGB contains infeasible ranges of color.



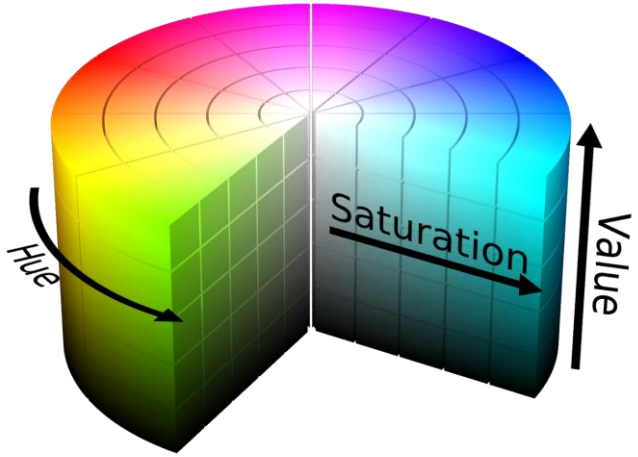
  - **YCbCr** (digital) and **YUV** (analog): Color spaces for video transmission and compression with *economic* bandwidth
    - The human visual system has higher spatial resolution for *luminance* than chrominance.

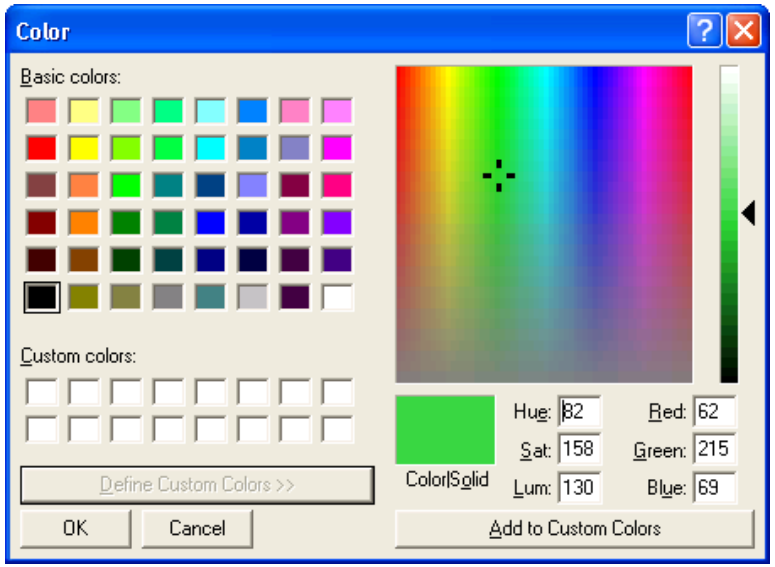
    → e.g. YCbCr (16-bit): *Y (8-bit)*, Cb (4-bit), Cr (4-bit)

Image: Wikipedia

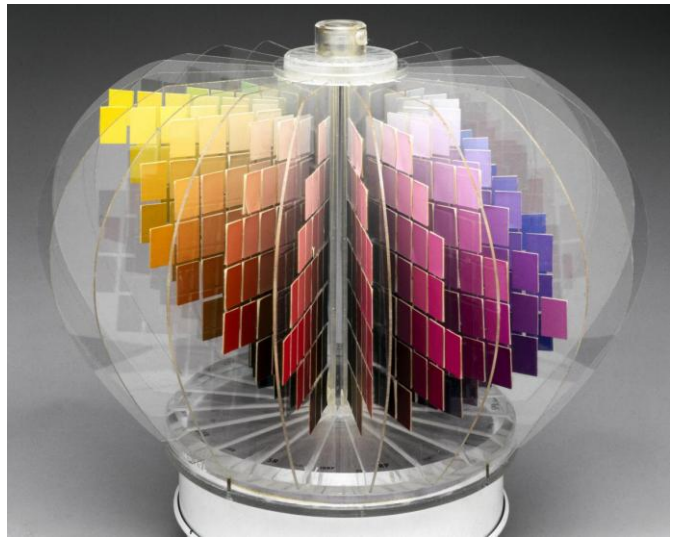
Color Spaces

- A **color space** is a specific system for representing color numerically and graphically. [show its list]
- Cylindrical-coordinate color spaces for **human-friendly color selection**
  - **HSV (HSL)**: Hue-Saturation-Value (색상-채도-명도 in Korean) designed by computer graphics researchers (1970s)
  - **Munsell**: Hue-Chroma-Value (색상-채도-명도 in Korean) used for paints, crayons, papers, soils, wires, … (since 1905)





Color Dialog Box

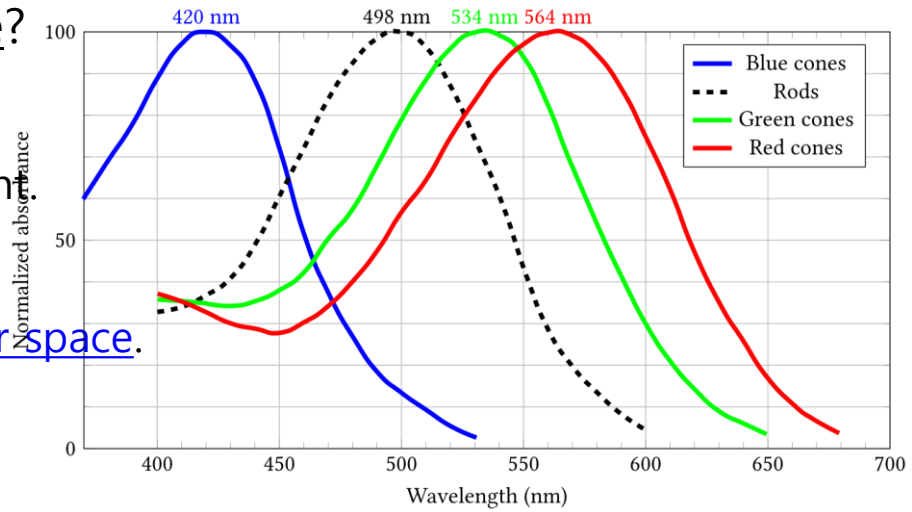


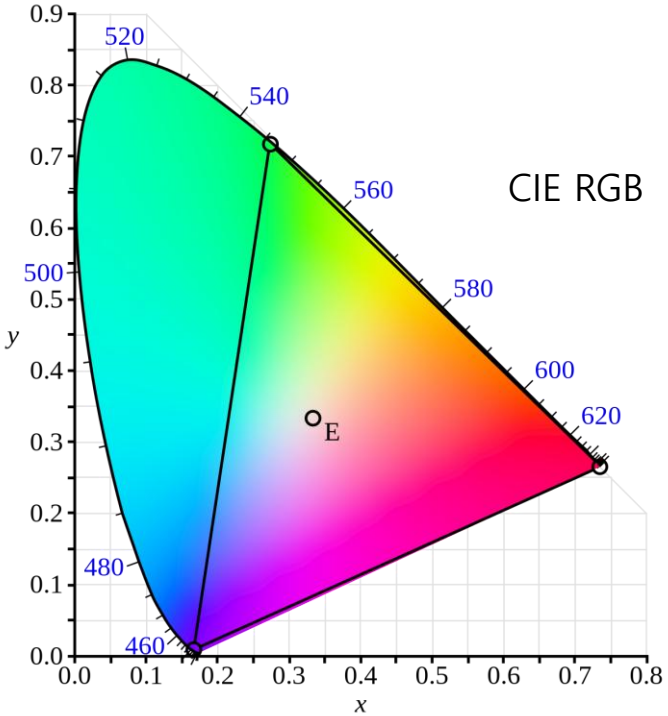
Munsell color tree

Image: Wikipedia, Microsoft, Works That Works

Color Spaces

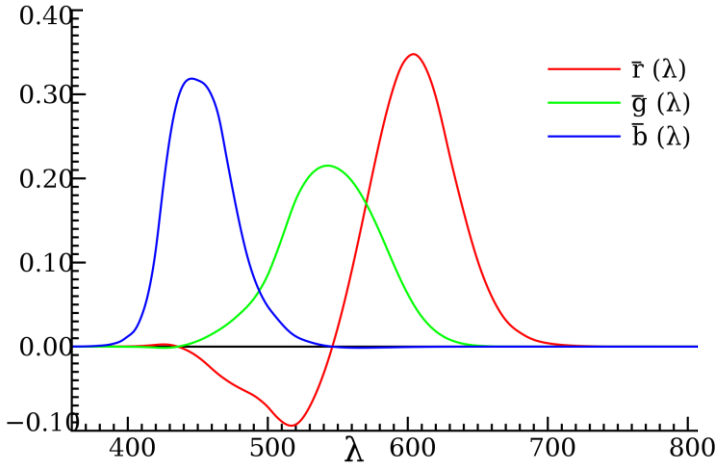
- A **color space** is a specific system for representing color numerically and graphically. [show its list]
- Q) Why are color spaces represented in the three-dimensional space?
  - Human eyes have three different cone cells (원추세포 in Korean).
    - They response long(L)-, medium(M)-, and short(S)-wavelength of light.
  - Human color vision is trichromatic.
  - The cone cell responses are modeled in LMS (long, medium, short) color space.





CIE RGB

CIE RGB color matching function



CIE XYZ standard human observer matching function

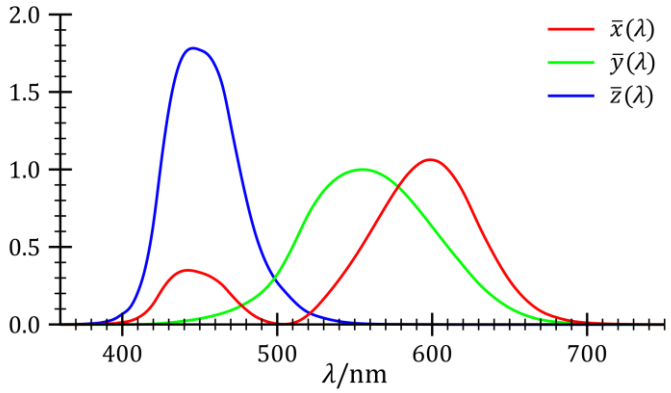


Image: Wikipedia

Color Spaces

- A **color space** is a specific system for representing color numerically and graphically. [show its list]
- Color spaces for **human visual perception**
  - **XYZ** (a.k.a. CIE 1931 XYZ): A standard color space based on the standard human observers
    - Note) Y ~ luminance (명암 in Korean)
    - The Hunt-Pointer-Estevez matrix (1980)

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 1.91020 & -1.11212 & 0.20191 \\ 0.37095 & 0.62905 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} L \\ M \\ S \end{bmatrix}$$

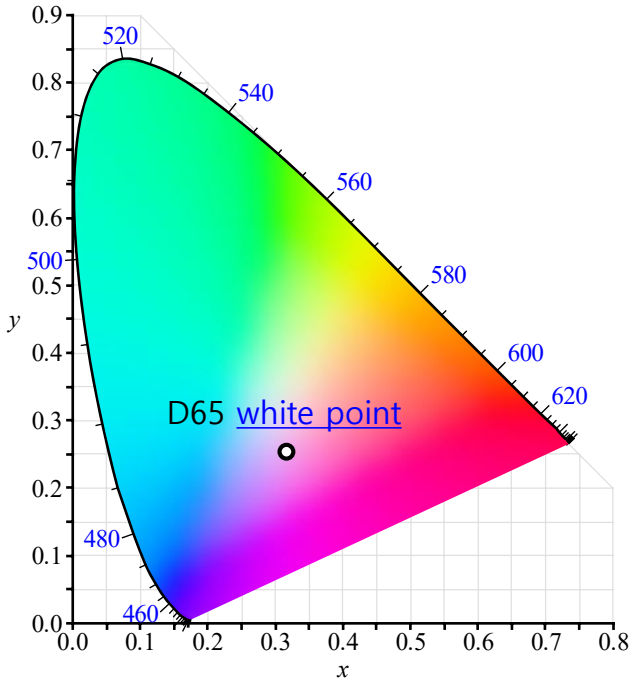
  - **xyY** (a.k.a. CIE xyY): A standard color space to represent the quality of color (regardless of its luminance)

$$x = \frac{X}{X+Y+Z}, \qquad y = \frac{Y}{X+Y+Z}, \qquad z = \frac{Z}{X+Y+Z} \qquad (x+y+z=1)$$

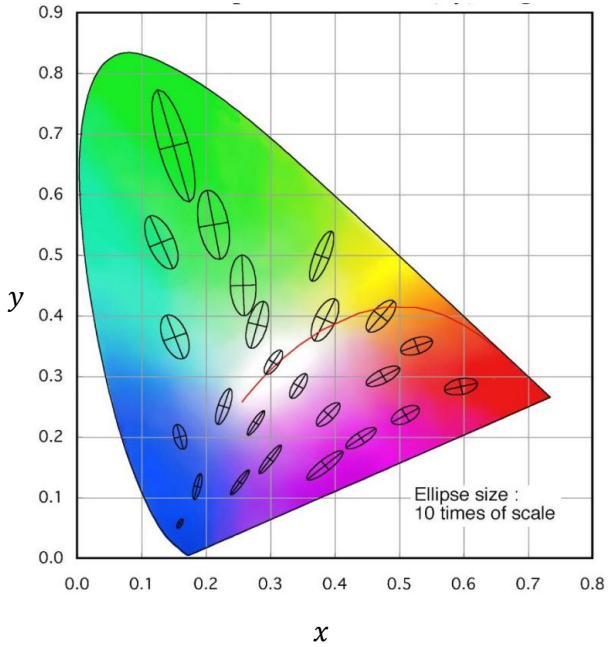
  - **LAB** (a.k.a CIE L*a*b*) and **LUV** (a.k.a. CIE 1976 L*u*v*): Standard color spaces for better representation

Color Spaces

- Color spaces for **human visual perception**
  - **XYZ** (a.k.a. CIE 1931 XYZ): A standard color space based on physiologically perceived colors by human observers
  - **xyY** (a.k.a. CIE xyY): A standard color space to represent the quality of color (regardless of its luminance)
  - **LAB** (a.k.a CIE L*a*b*) and **LUV** (a.k.a. CIE 1976 L*u*v*): Standard color spaces for better representation
    - **CIE xy chromaticity diagram**



CIE 1931 with MacAdam ellipse



CIE 1976 with MacAdam ellipse

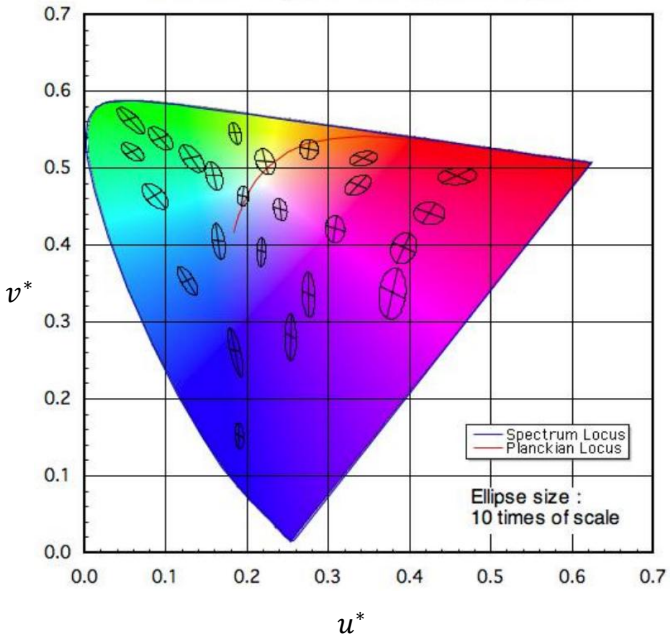


Image: Wikipedia, Yoshi Ohno

Color Spaces

- Example) Color space conversion (HSV) [color_bgr2hsv.py]

```
import numpy as np
import cv2 as cv

img = cv.imread('../data/peppers.tif')
assert img is not None, 'Cannot read the given image'

# Convert the BGR image to its HSV image
img_hsv = cv.cvtColor(img, cv.COLOR_BGR2HSV)

# Show hue, saturation, and value channels as color images
img_hue = np.dstack((img_hsv[:,:,0],
                     np.full_like(img_hsv[:,:,0], 255),
                     np.full_like(img_hsv[:,:,0], 255)))
img_hue = cv.cvtColor(img_hue, cv.COLOR_HSV2BGR)
img_sat = np.dstack((img_hsv[:,:,1], ) * 3)
img_val = np.dstack((img_hsv[:,:,2], ) * 3)
merge = np.hstack((img, img_hue, img_sat, img_val))
cv.imshow('Color Conversion: Image | Hue | Saturation | Value', merge)
cv.waitKey()
cv.destroyAllWindows()
```

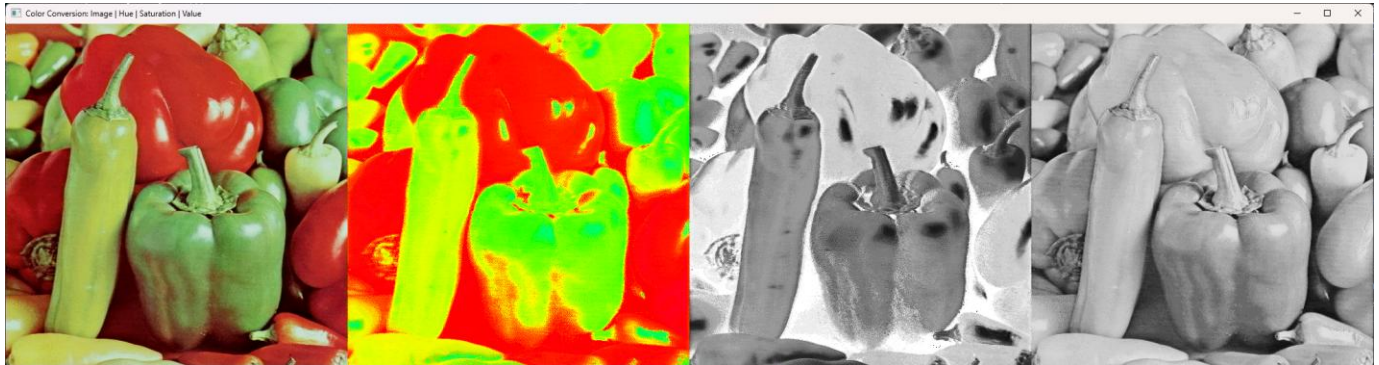
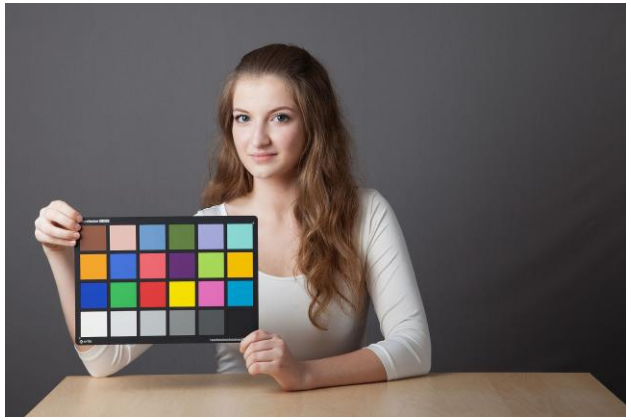


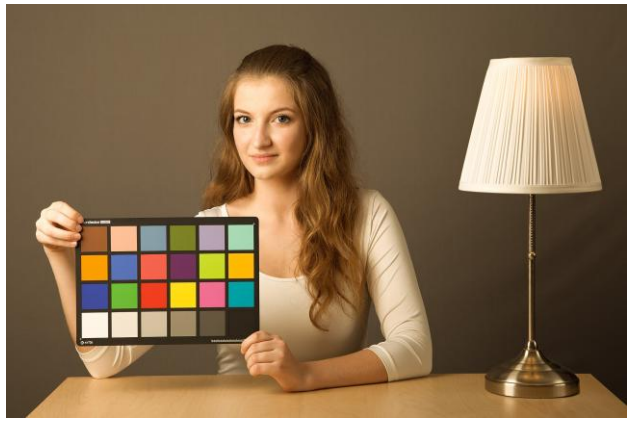
Image: Wikipedia (color balance, color temperature)

Color Image Processing

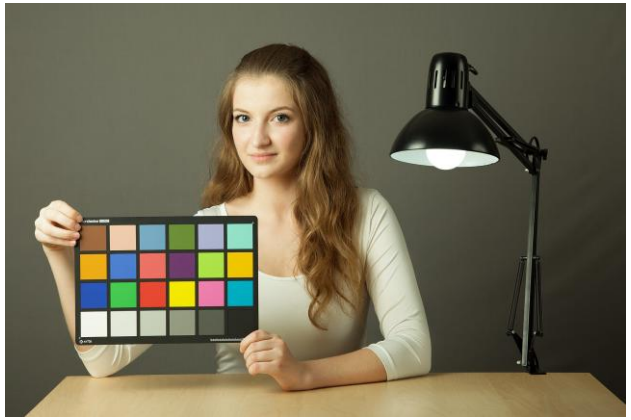
- **Color balancing** (or color constancy) is a global adjustment of the intensities of the color (typically RGB)
  - **White balance** (shortly WB): A common color balancing method that makes a white object appear white under different illuminations



Neutral light

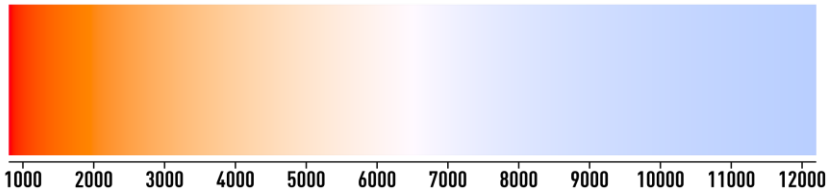


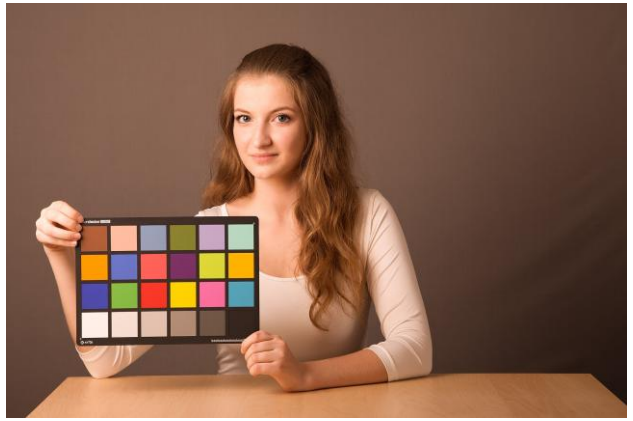
Warm (reddish) light

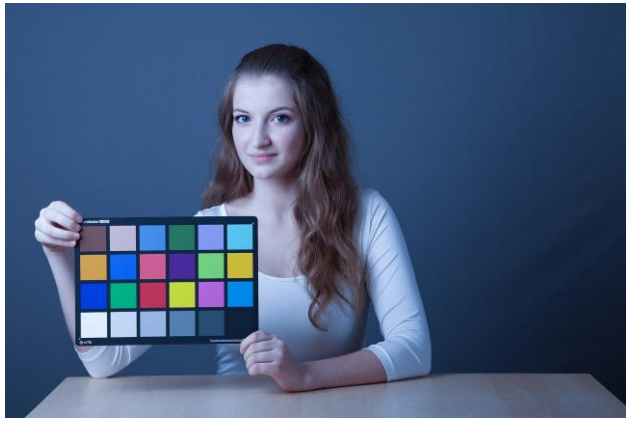


Cold (bluish) light

  - Note) **Color temperature**: A parameter describing the color of light source (as temperature; unit: [K])







Color Image Processing

- **Color segmentation** is the process of partitioning an image by grouping pixels based color properties.
  - e.g. Colored objects, road lanes, …, skin

  - Q) How can we represent a color range? (What is a good color space?)
    - Thresholds (~ box), an ellipse, Gaussian mixture model (GMM; ~ a set of ellipses), …
- **Color histogram** (as a descriptor)
  - e.g. Person re-identification (ReID)

Image: [Color Segmentation using GMM](), [Pytorch ReID]()

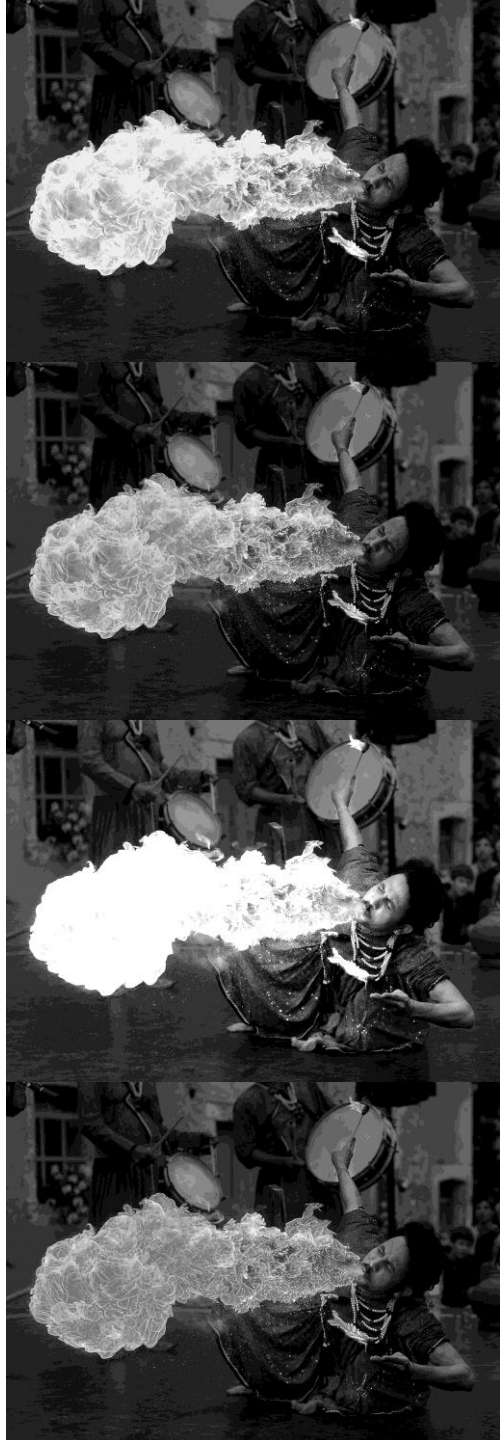
Color Image Processing

- Many classical algorithms only assume **gray-scale images**, not color images.
  - e.g. Histogram equalization, edge detection, …

```
# Read the given image as gray scale
img = cv.imread('../data/lena.tif', cv.IMREAD_GRAYSCALE)
```

  - Why? (Difficulty in dealing with three-dimensional values)
    - Three-dimensional comparison? (e.g. [0, 75, 0] vs. [25, 25, 25])
  - Ad-hoc solutions to use the classical algorithms (only for gray-scale images)
    - ~~Idea #1) Apply the algorithms to each RGB channel of input images~~
    - Idea #2) Apply the algorithms only to **the gray-channel of input images**
      - e.g. RGB → **Y**CbCr → RGB
      - Note) However, the gray-channel is ambiguous.

Color



L*a*b*

I (Intensity)

HS**V**

HS**L**

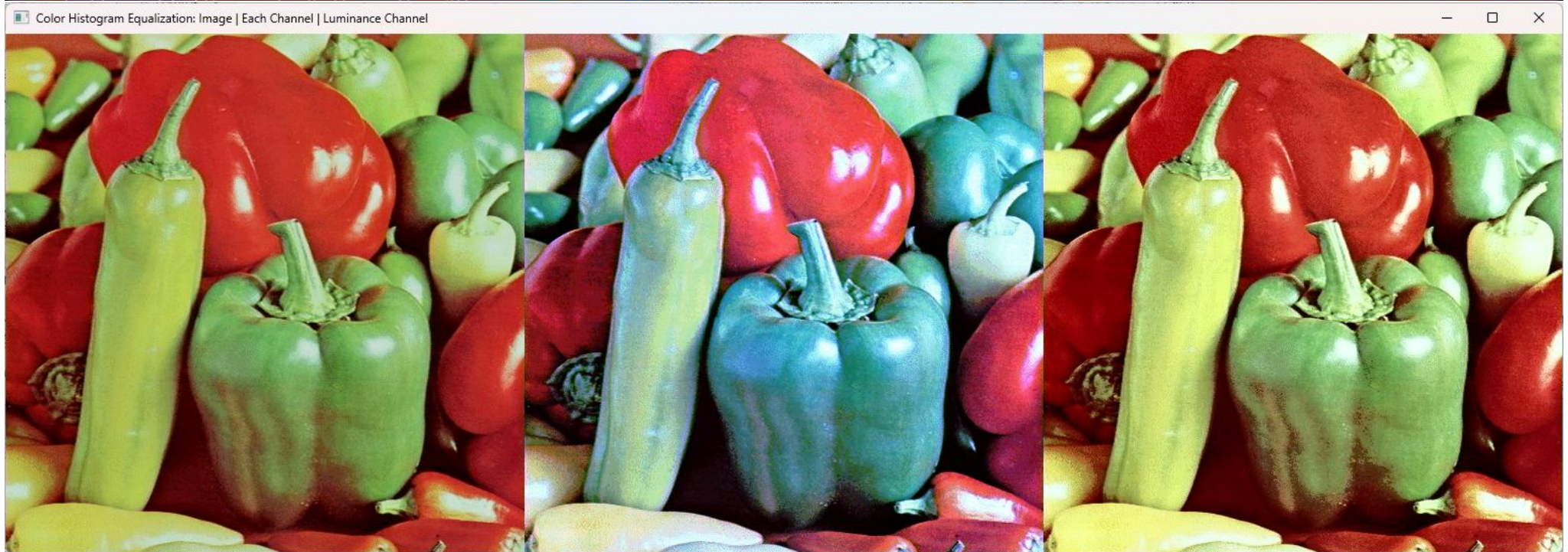
Image: [Wikipedia]

Color Image Processing

- Example) Color histogram equalization [histogram_equalization+color.py]

```
img_list = [...]

# Initialize a control parameter
img_select = 0

while True:
    # Read the given image
    img = cv.imread(img_list[img_select])

    # Apply histogram equalization to each channel
    img_hist1 = np.dstack((cv.equalizeHist(img[:,:,0]),
                           cv.equalizeHist(img[:,:,1]),
                           cv.equalizeHist(img[:,:,2])))

    # Apply histogram equalization only to the luminance channel in YCbCr
    img_cvt = cv.cvtColor(img, cv.COLOR_BGR2YCrCb)
    img_hist2 = np.dstack((cv.equalizeHist(img_cvt[:,:,0]),
                           img_cvt[:,:,1],
                           img_cvt[:,:,2]))
    img_hist2 = cv.cvtColor(img_hist2, cv.COLOR_YCrCb2BGR)

    # Show all images
    merge = np.hstack((img, img_hist1, img_hist2))
    cv.imshow('Color Histogram Equalization: Image | Each Channel | Luminance Channel', merge)
    key = cv.waitKey()
    if key == 27: # ESC
```

Color Histogram Equalization: Image | Each Channel | Luminance Channel


Color Histogram Equalization: Image | Each Channel | Luminance Channel

Summary

- **Photometric Image Formation**
  - CMOS sensor: Light intensity sensor
  - Color camera = **Bayer filter** + CCD/CMOS sensor
- **Color Spaces**
  - Color spaces for **media**: RGB, CMY(K), YCbCr, and YUV
  - Color spaces for **human-friendly color selection**: HSV, HSL, and Munsell
  - Color spaces for **human visual perception**: XYZ, xzY, LAB (L*a*b*), and LUV (L*u*v*)
  - Example) Color space conversion
- **Color Image Processing**
  - Color balancing (e.g. white balance, color temperature, …)
  - Color segmentation (e.g. detecting colored objects, …)
  - Color histogram as a descriptor (e.g. person re-identification, …)
  - Tip) How to use the classical algorithms only support gray-scale images
    - Example) Color histogram equalization